INCH-POUND
MIL-PRF-49450/13(CR)
10 March 2000

PERFORMANCE SPECIFICATION SHEET

BATTERY, RECHARGEABLE, NICKEL-CADMIUM, VENTED, AIRCRAFT, BB-649A/A

This specification is approved for use by the USACECOM, Department of Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-49450(CR).

Performance.

Voltage. The voltage of the battery shall be 24 Vdc (nominal).

Capacity. The battery capacity shall be 18 ampere hours.

Weight. The weight of the battery shall be 49 pounds (Max.)

Connector(J1) The main power connector shall be an quick disconnect type that meets the dimensional and interface requirements MS3509.

Cells. The battery shall be constructed using 19 BB-648A/A cells.

DRAWING NOTES:

All dimensions are in inches, unless specified otherwise.

Vent Pipes The battery shall have two vent pipes shown in figure 1 and figure 2. The vent pipes shall be of the same material as the battery case. They shall be .813± 062 inches long with a diameter of 0.608 ±.008 inches.

Holding strap The holding strap shown in figure 2 and figure 3 shall be placed on the centerline of the battery.

AMSC N/A: FSC 6140

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

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Main power receptacle (J1). The battery shall utilize a two-pin receptacle conforming to the interface requirements of MS3509. The J1 receptacle shall be centered as shown in figure 1.

Battery Container: The body and cover shall be made of the same alkali-resistant material. The dimensions of the container and the dimensions and locations of the receptacles mounting holes and hold down strap shall conform to figures 1, 2 and 3.

Color. The color of the container and cover except latches, hold-down hooks other external hardware, pads, and identification markings and identification shall conform to FED-STD-595 within the range of blue color chips 15090 to 15193.

REQUIREMENTS:

Nomenclature:	BB-649A/A Battery, Rechargeable
Rated capacity:	18 ampere-hours
Voltage (Nominal):	24 Volts
Weight (Maximum):	49 pounds
Connector:	one 2 pin connector that meets the dimensional and interface requirements of MS3509(J1)
Cells:	19, BB-648A/A

Test	Discharge Rate(Amps)	Cut Off Voltage(V)	Capacity(AH) Minimum
Capacity	12	18	18
Cycle Life			
Normal Rate	18	18	16.2
High Rate	180	14.4	12
Vibration	18	18	18
Temperature Shock	18	18	18
Altitude	18	18	15.3
Humidity and Charge Retention	18	18	13.5
High Rate			
Discharge and Low Temperature	180	13	9
High Rate			
Discharge and Operating Position	180	14.4	15.3
Shock	18	18	18

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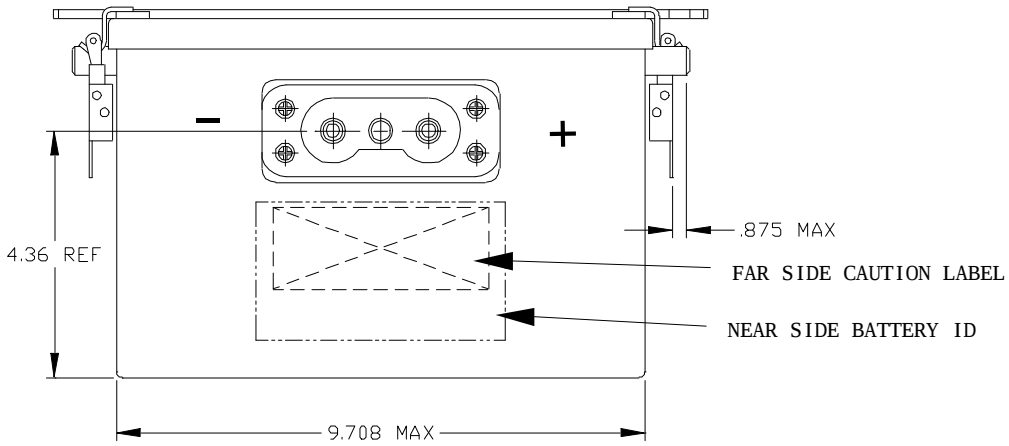

FIGURE 1: BATTERY RECHARGEABLE BB-649A/A FRONT VIEW

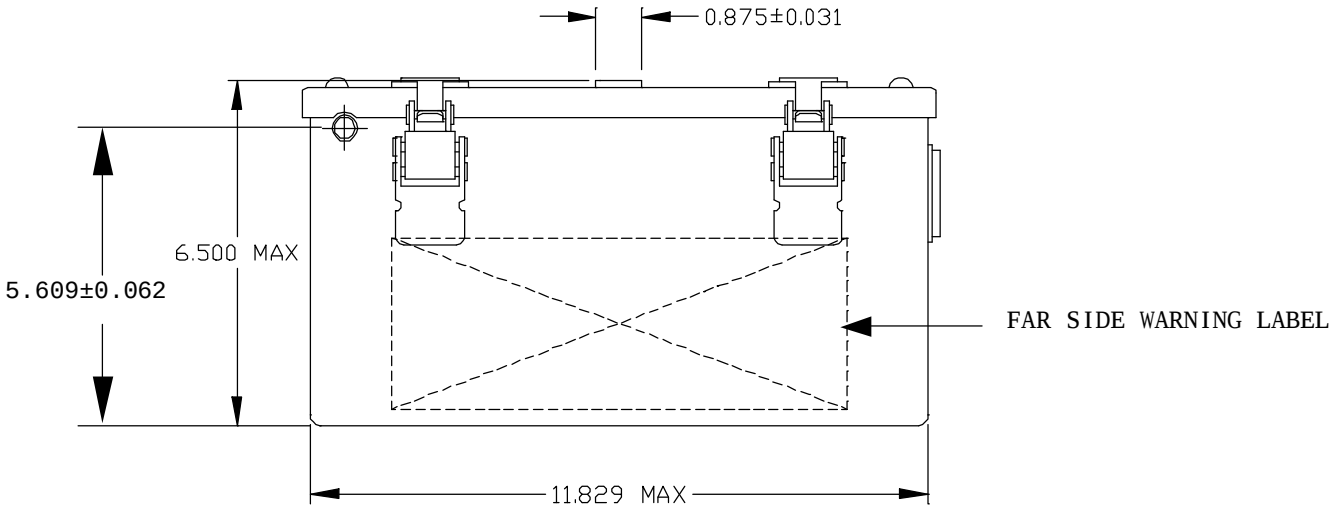

FIGURE 2: BATTERY RECHARGEABLE BB-649A/A SIDE VIEW

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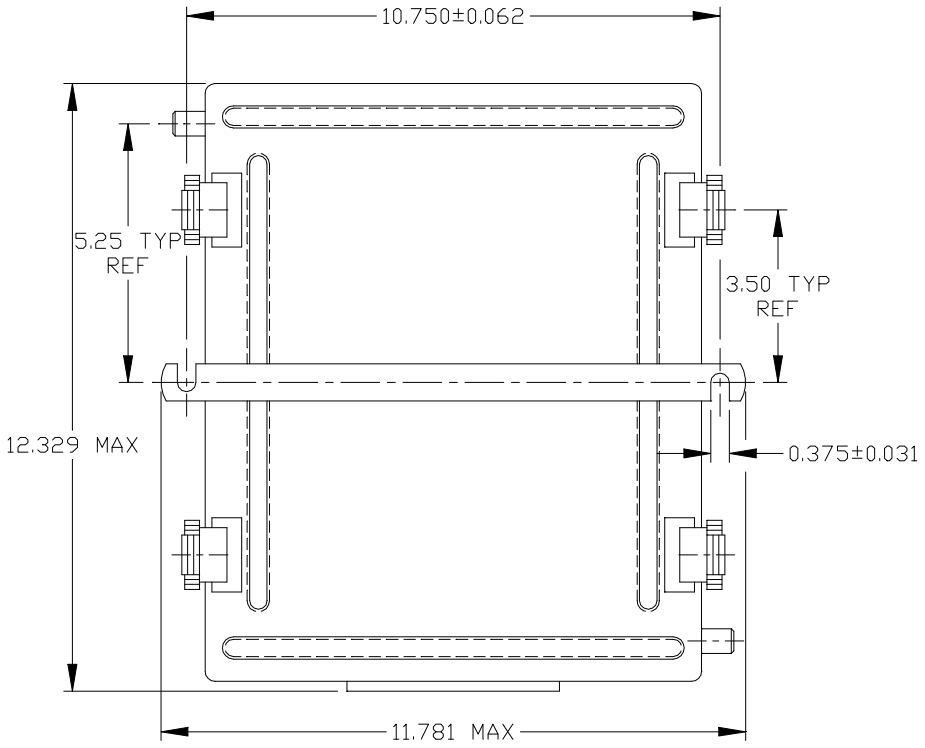

FIGURE 3: BATTERY RECHARGEABLE BB-649A/A TOP VIEW

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